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📘 **AI/ML Internship – Day 12 Notes & Tasks**

🟢 **Module 3: Python for AI**

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📌 **Part 1: List Comprehension (Very Important)**

🔹 **What is List Comprehension?**

A short and efficient way to create lists.

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🔹 **Basic Syntax**

[expression for item in iterable]

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🔹 **Example**

```
nums = [1, 2, 3, 4]

squares = [x*x for x in nums]
print(squares)
```

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🔹 **With Condition**

```
nums = [1, 2, 3, 4, 5]

even = [x for x in nums if x % 2 == 0]
print(even)
```

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🔹 **Real Use**

```
names = ["ai", "ml", "python"]

upper = [n.upper() for n in names]
print(upper)
```

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📌 **Part 2: Dictionary Comprehension**

🔹 **What is it?**

Create dictionary in one line.

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🔹 **Example**

```
nums = [1, 2, 3]

squares = {x: x*x for x in nums}
print(squares)
```

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🔹 **With Condition**

```
nums = [1, 2, 3, 4]

even_dict = {x: x*x for x in nums if x % 2 == 0}
print(even_dict)
```

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📌 **Part 3: Sets (Advanced Usage)**

🔹 **What is Set?**

- Unordered collection
- No duplicates

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🔹 **Example**

```
nums = {1, 2, 2, 3}
print(nums)
```

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🔹 **Operations**

```
a = {1, 2, 3}
b = {3, 4, 5}

print(a | b)  # Union
print(a & b)  # Intersection
print(a - b)  # Difference
```

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## 📌 Part 4: Iterators & Generators

### 🔹 Iterator

Object that can be iterated.

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### 🔹 Generator (Important)

👉 Uses yield instead of return

```
def count(n):
    for i in range(n):
        yield i

for num in count(5):
    print(num)
```

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### 🔹 Why Generators?

- Memory efficient
- Used in large data processing

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## 📌 Part 5: Zip Function

### 🔹 Combine Multiple Lists

```
names = ["A", "B"]
marks = [80, 90]

combined = list(zip(names, marks))
print(combined)
```

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## 📌 Part 6: Enumerate Function

```
names = ["AI", "ML"]

for i, name in enumerate(names):
    print(i, name)
```

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📌 **Part 7: Real-World Thinking**

👉 These concepts are used in:

- Data transformation
- Feature engineering
- Data pipelines

👉 Especially:

- List comprehension → fast processing
- Generators → big data handling

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🎯 **Tasks for Day 12**

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🧠 **Task 1: Theory**

1. What is list comprehension?
2. Difference between list and set
3. What is generator?
4. What is zip function?
5. What is enumerate?

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💻 **Task 2: List Comprehension**

1. Square numbers
2. Find even numbers
3. Convert strings to uppercase
4. Filter numbers > 50

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💻 **Task 3: Dictionary Comprehension**

1. Create number-square dictionary

2. Filter even numbers
3. Map names to lengths

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💻 **Task 4: Set Practice**

1. Remove duplicates
2. Find common elements
3. Find union and difference

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💻 **Task 5: Generator Practice**

1. Create generator for numbers
2. Generate even numbers
3. Fibonacci using generator

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💻 **Task 6: Zip & Enumerate**

1. Combine 2 lists
2. Print index + value
3. Convert zip result to dictionary